THE FUNDAMENTAL RELATIVE CLASSES COMPOSE ALONG A FLAT COMPOSITION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Composition of closed subscheme cycle families (relative-cycles chapter, Lemma lemma-construction-closed, canonical tag 0H6M). cited at proof line 3073. Specialized to Z = X and W = Y: the composition of the fundamental classes is the family of the fibred product, which is X itself.

Source reference. Original lemma 0H6M: sources/stacks-chow/tags.txt, source lines 21427–21427. Cited at proof line 3073: the special case Z = X, W = Y.

Lemma 0.2. *Let $f : X \to Y$ and $g : Y \to S$ be a morphisms of schemes. Assume S locally Noetherian, g locally of finite type and flat of relative dimension $e \geq 0$, and f locally of finite type and flat of relative dimension $r \geq 0$. Then $[X/X/Y]_r \circ [Y/Y/S]_e = [X/X/S]_{r+e}$ in $z(X/S, r+e)$.*

Proof. Special case of Lemma 0H6M. □

Date: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.